
MAM3000W 3AL

University of Cape Town

Final Examination June 2022

Department of Mathematics and Applied Mathematics

Time: 120 minutes

Available marks: 110

Maximum marks: 100

Instructions

1. Answer all questions in the lined answer booklets.
2. **Start each question on a new page. Number each page and put your student number on the top of each page.**
3. Show all work and reasoning.
4. This exam consists of **5** questions. Make sure your question paper is complete.
5. If you've read the instructions write the name of your favourite mathematician on the front of your booklet (don't write it where *your* name is supposed to go) +1 mark

Good luck! :)

1. (a) State the First Isomorphism Theorem.
(b) Let G be an abelian group. Fix a positive integer n . We define two subgroups:
 $A = \{g \in G : g^n = 1\}$ and $B = \{g^n : g \in G\}$. Prove that $G/A \cong B$.

[5 + 10 = 15 marks]

2. (a) State the Orbit Decomposition Theorem.
(b) Prove that if G is a p -group acting on a set X , then $|X| \equiv |X_f| \pmod{p}$.
(c) Let G be a group and H a normal subgroup of G .
(i) Show that conjugation by G on H is a group action.
(ii) Let p be a prime. Prove that if G is a p -group, then $H \cap Z(G)$ is nontrivial.

[5 + 5 + 5 + 10 = 25 marks]

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3. (a) State the definition of a normaliser.
- (b) State the Second Sylow Theorem.
- (c) Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Let H and K be Sylow p -subgroups of G .
- (i) Prove that if $K = aHa^{-1}$ for some $a \in G$, then $H = a^{-1}Ka$.
- (ii) Prove that $N_G(H)$ and $N_G(K)$ are conjugates.

[5 + 5 + 5 + 10 = 25 marks]

4. (a) State the Third Sylow Theorem.
- (b) Prove the Third Sylow Theorem, omitting the proof that $n_p \mid m$.

[5 + 15 = 20 marks]

5. (a) State the definition of a simple group and give an example.
- (b) Show that any group of order $520 = 2^3 \times 5 \times 13$ is not simple.

NOTE: The divisors of 104 are 1, 2, 4, 8, 13, 26, 52, 104. Moreover, $4 \times 26 = 104$ and $12 \times 40 = 480$.

[5 + 20 = 25 marks]